

# Daksh Dheer

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## Interests

Motivated to work in algebraic number theory, particularly class groups, and related areas, with strong background in algebra, representation theory (of quivers and finite groups), homological algebra, valuation theory, algebraic topology, and commutative algebra.

## Education

- Master of Science (by Research) Mathematics (2026) : CGPA: 9.55  
*Indian Institute of Science Education and Research (IISER), Thiruvananthapuram*  
**Batch topper**
- Bachelor of Science (Honours) Mathematics (2023) : CGPA : 9.23  
*Hansraj College, University of Delhi*
- CBSE XII (2020) : 97.5%  
*Ryan International School, Jaipur*

## Experience

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| August 2025 – April 2026     | <b>Teaching Assistant</b><br>Teaching tutorial classes, conducting exams as well as correcting papers for two bachelor's level courses – Introduction to Probability, and Single Variable Calculus.                                                                                                                                                                                                       |
| April 2025 – May 2026        | <b>Master's Thesis</b><br><i>Under the supervision of Prof. Viji Z. Thomas, IISER Thiruvananthapuram</i><br><b>Title: <i>On the General Theory of Valuations and Class Field Theory</i></b> - currently reading Neukirch's Algebraic Number Theory (chapter 2), after having covered the first three chapters from Valued Fields by Engler and Prestel. Aiming to cover class field theory by graduation. |
| February 2024 – October 2024 | <b>Guided Reading Project – Report</b><br><i>Under the supervision of Prof. Viji Z. Thomas, IISER Thiruvananthapuram</i><br>Read several sections and chapters from M. Isaacs': Finite Group Theory and J.P. Serre's Finite Groups, particularly concerning the transfer homomorphism, advanced properties of $p$ -Sylow subgroups and structure of subgroups of products of groups.                      |
| Jan 2021 – April 2023        | <b>College Societies</b><br><i>The Poetry Society</i> (January 2021 – April 2023) Served as a member of the Editorial and Technical teams of Kavyanjali, the poetry society of Hansraj College.<br><br><i>Debating Society</i> (January 2021 – August 2021) Participated in debates and improved communication skills while being part of the English Debating society.                                   |

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| Oct 2020 – Aug 2022 | <b>Editor - Undergraduate Academic Journal (Aankalan) – Articles</b><br><i>Editor-in-Chief:</i> (Aug 2021 – Aug 2022) Served as the Ed-in-Chief of the third edition of the Annual Academic Journal of the Mathematics Department, Hansraj College.<br><i>Assistant Editor:</i> (October 2020 – August 2021) Served as an assistant editor in the second edition of the journal. |
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## Research Schools and Conferences Attended

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| January 2026          | <i>Frontier Symposium in Mathematics 2026, conducted by Indian Institute of Science Education and Research (IISER) Thiruvananthapuram</i>                                                 |
| October 2025          | <i>International Conference on Class Groups of Number Fields and Related Topics (ICCGNFRT - 2025), hosted at SRM University AP</i>                                                        |
| June 2025             | <i>Mini course on Spectral Sequences and their applications in Homological Algebra (2025), conducted by Indian Institute of Science Education and Research (IISER) Thiruvananthapuram</i> |
| January 2025          | <i>Frontier Symposium in Mathematics 2025, conducted by Indian Institute of Science Education and Research (IISER) Thiruvananthapuram</i>                                                 |
| December 2024         | <i>Representation Theory of Quivers and BGG Category O for semi simple Lie Algebras (2024), conducted by Indian Institute of Technology (IIT) Kanpur</i>                                  |
| June 2024 - July 2024 | <i>Annual Foundation School - III, conducted by Indian Institute of Space and Technology (IIST) Thiruvananthapuram</i>                                                                    |
| February 2024         | <i>Frontier Symposium in Mathematics 2024, conducted by Indian Institute of Science Education and Research (IISER) Thiruvananthapuram</i>                                                 |
| July 2022             | <i>Madhava Nurture Camp'22 - conducted by Indian Institute of Technology Bhubaneswar</i>                                                                                                  |
| November 2021         | <i>Madhava Nurture Camp'21 - conducted by St. Xavier's College, Kolkata</i>                                                                                                               |

## Awards and Recognition

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- Batch topper, MS (by Research) Mathematics (2023 – 2026).
- **INSPIRE - Scholarship for Higher Education** fellowship, 2020-2023.
- Delhi University topper, B.Sc. (Hons.) Mathematics (2021).
- **Special academic prizes awarded (undergraduate):** Shrimati Krishnawanti Daulat Ram Chadha Prize, Shri Ladha Ram Delory Prize, Alumunus Prof. Nem Kumar Jain Prize, Pradeep Kumar Gupta Prize.

## Skills

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| Critical Thinking | MATLAB                          |
| Communication     | Python                          |
| Collaboration     | L <sup>A</sup> T <sub>E</sub> X |

## Relevant Coursework

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Some of the courses and topics I have studied as part of my coursework:

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|--------------------------------|-------------------------------------------|------------------------------------------|---------------------------|
| • Discrete Mathematics         | • Complex Analysis                        | • Homological Algebra                    | • Differential Geometry   |
| • Groups and Rings             | • Algebraic Topology                      | • Commutative Algebra                    | • Category Theory         |
| • Fields, Modules and Algebras | • Number Theory and Cryptography          | • Finite and Infinite Galois Theory      | • Algebraic Number Theory |
| • Analysis on Manifolds        | • Hyperbolic Geometry and Fuchsian Groups | • Representation Theory of Finite Groups | • General Topology        |

## References

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**Prof. Viji Z. Thomas**  
*Thesis Advisor*  
[vthomas@iisertvm.ac.in](mailto:vthomas@iisertvm.ac.in)

**Dr. Shrihari Sridharan**  
School of Mathematics  
[shrihari@iisertvm.ac.in](mailto:shrihari@iisertvm.ac.in)